

Problem-Solution-Fit

Date	29 June 2026
Team ID	xxxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

Section	Details
1. CUSTOMER SEGMENT(S) [CS]	• Farmers (Primary Users) • Agricultural Researchers • Government & Policymakers • Agricultural Extension Officers
2. PROBLEMS / PAINS + ITS FREQUENCY [PR]	• Difficulty in selecting the right crop (Very Frequent) • Low crop yield due to unsuitable soil and climate (Frequent) • Inefficient use of fertilizers and water (Frequent)
3. TRIGGERS TO ACT [TR]	• Need to increase crop yield • Seasonal crop planning • Changing weather conditions • Rising farming costs
4. EMOTIONS BEFORE / AFTER [EM]	Before: Confused, Worried, Uncertain After: Confident, Satisfied, Optimistic
5. AVAILABLE SOLUTIONS – PROS & CONS [AS]	Available Solutions: Traditional farming advice, agricultural experts, weather apps. Pros: Easy access, experienced guidance. Cons: Time-consuming, not personalized, less accurate, manual decision-making.

6. CUSTOMER LIMITATIONS [CL]	• Limited budget • Limited internet connectivity in rural areas • Lack of technical knowledge • Limited access to smart devices
7. BEHAVIOR + ITS INTENSITY [BE]	• Consults local experts before farming (High) • Checks weather forecasts (Medium) • Relies on previous farming experience (High)
8. CHANNELS OF BEHAVIOR [CH]	Online: Web Application, Agricultural Websites, Weather Services Offline: Agricultural Officers, Local Markets, Farmer Meetings
9. PROBLEM ROOT / CAUSE [RC]	• Lack of accurate crop prediction tools • Changing environmental conditions • Limited awareness of soil nutrient analysis • Dependence on traditional farming practices
10. YOUR SOLUTION [SL]	OptiCrop uses Machine Learning to analyze soil nutrients (N, P, K), pH, temperature, humidity, and rainfall to recommend the most suitable crop. It helps farmers improve productivity, optimize resources, and support sustainable agriculture.